Machine Learning for CO_2 Emission Optimization in Delivery Logistics

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Abstract

Delivery logistics must balance emissions, travel time and operational constraints. This report investigates whether a machine-learning model trained on historical delivery records can identify lower-emission configurations without substantially increasing travel time. Using a dataset of 100,000 deliveries supplied for this project, we train and compare five regression models to predict CO_2 emissions from pre-dispatch features. The best model (Ridge regression with interaction terms) achieves near-perfect predictive accuracy on held-out data (test $R^2 = 1.000$, RMSE $= 4.26 \times 10^{-6}$). When this model is used to search over vehicle type, road type, departure hour and planned speed under a 10% travel-time allowance, no configuration yields a meaningful reduction in predicted emissions: the optimiser leaves every one of 400 test deliveries unchanged. A second, separate analysis examines hypothetical lower-traffic route alternatives and derives concrete break-even detours. The near-perfect predictions are evidence of a simple underlying data-generating formula rather than evidence that the same model would predict real-world emissions perfectly. The work also highlights important limitations, including unknown units, missing operational constraints, and the risk of an optimiser exploiting residual model error. Overall, the results show that high predictive accuracy alone does not guarantee actionable emission savings when the tested decision variables offer no meaningful improvement within the dataset.

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1 Introduction and Problem Formulation

Road freight is a major source of greenhouse-gas emissions. Even modest improvements in how deliveries are planned-choice of vehicle, route characteristics, departure time and planned speed-can therefore have a measurable environmental impact, provided those choices actually affect emissions [1, 2].

In practice a dispatcher faces a constrained decision problem. For a given shipment (origin, destination, cargo weight) the planner may vary vehicle type, road type, departure hour and planned average speed. Any change must keep travel time within an acceptable margin of the historically observed time, otherwise the service level is compromised.

The research question of this project is therefore:

> *Can a machine-learning model help identify delivery configurations that lower predicted CO_2 emissions without significantly increasing travel time?*

The approach has two stages. First a regression model is trained to predict emissions from features that are known before the vehicle leaves the depot ("pre-dispatch" features). Second, the trained model is used as a surrogate objective inside a constrained search over the decision variables listed above. Formally, for each delivery i with fixed attributes a_i we solve

$$\min_{d \in \mathcal{D}} \hat{c}(a_i, d) \quad \text{subject to} \quad T(a_i, d) \leq (1 + s)\, T_i^{\text{base}}, \tag{1}$$

where $\hat{c}$ is the model's emission prediction, T is the implied travel time, T_i^{base} is the travel time recorded in the data, and s is a relative slack (default $s = 0.10$).

A reduction found by this procedure is only a *predicted* reduction. Whether the same reduction would materialise in the real world depends on the quality of the data and of the model-a point examined carefully in later sections.

2 Dataset and Exploratory Data Analysis

2.1 Data overview

The file `logistics_co2_dataset.csv`, supplied for this project, contains 100,000 delivery records and 19 columns. There are no missing values and no duplicate rows. The variables fall into three groups:

- **Geography and route**: origin and destination coordinates, distance (km), elevation gain and loss, road type (coded 0/1/2).
- **Operational**: cargo weight, vehicle type (coded 0/1/2), number of stops, planned average speed, departure hour (0–23).
- **Environment and outcome**: ambient temperature, rain indicator, traffic index (0–1), CO_2 emission (target), travel time, delay risk.

Summary statistics match the figures supplied with the assignment (distance mean ≈ 1007 km, CO_2 mean ≈ 315, cargo weight between 1 and 30, etc.). Units for emissions, cargo and travel time are not documented.

2.2 Distributions

Figure 1 shows the marginal distributions of CO_2 emissions and route distance on the training split. Route distances are spread almost uniformly between about 10 and 2,000 km (skewness near zero), which is unusual for real delivery data. CO_2 emissions are right-skewed with a long tail of high-emission trips; this skew is inherited from the combination of distance and cargo rather than from isolated anomalies.

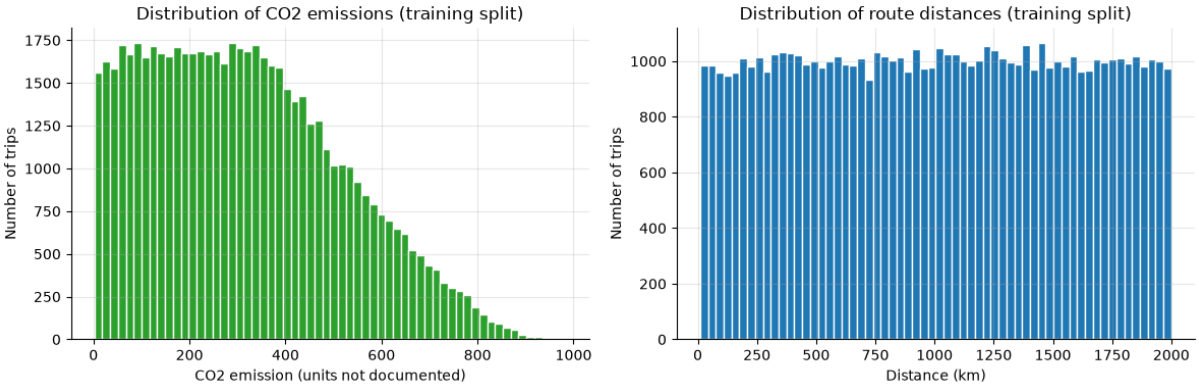


Figure 1: Distribution of CO_2 emissions (left) and route distances (right) on the training split.

2.3 Associations with the target

Pearson correlations with `co2_emission` (training split, pre-dispatch features only) are dominated by three variables:

Variable	Pearson r
`distance_km`	0.90
`cargo_weight`	0.37
`traffic_index`	0.07

Vehicle-type and road-type group means of CO_2 differ by at most 1.34 and 1.02 units respectively-differences that are negligible relative to the scale of the target (standard deviation $\approx$ 200).

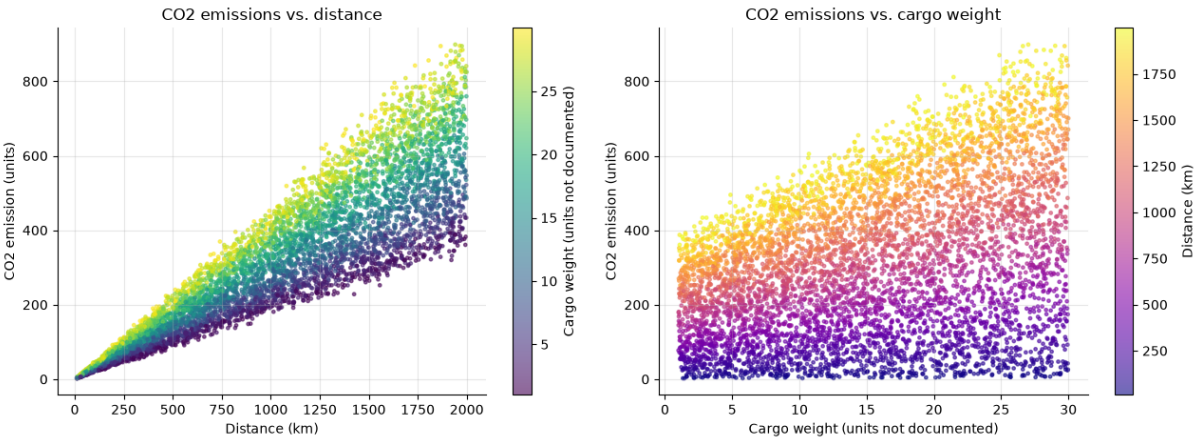


Figure 2: CO_2 emissions versus distance (left, coloured by cargo weight) and versus cargo weight (right, coloured by distance). Sample of 6,000 training observations.

Figure 2 visualises the dominant relationships. Emissions grow with distance, and at a given distance heavier cargo produces higher emissions.

2.4 Key data-quality findings

Three consistency checks are decisive:

1. Travel time obeys the exact identity

$$\texttt{travel_time} = \frac{\texttt{distance_km}}{\texttt{avg_speed}} \times 60 \times (1 + \texttt{traffic_index})$$

   with a maximum relative error of 7.3×10^{-16}. Consequently travel time is a deterministic function of distance, speed and traffic.

2. The Euclidean (great-circle) distance computed from the latitude/longitude pairs is essentially uncorrelated with the recorded `distance_km` ($r = -0.004$). In 72 % of records the recorded route distance is even shorter than the straight-line distance, which is physically impossible for a real road network.

3. Once interaction terms of the form `distance`×`cargo`, `distance`×`traffic` and `distance`×`elevation` are included, a linear model reproduces the target with a maximum absolute error of 6×10^{-11} on the test set.

The analysis revealed two important findings. First, CO_2 emissions can be predicted with very high accuracy using a small set of features. However, the recorded distances do not closely match distances calculated from the geographic coordinates. This raises questions about how well the dataset represents real delivery operations. Therefore, the results should be interpreted with caution, especially when considering their application to real-world logistics.

3 Data Preparation and Feature Engineering

The dataset contains no missing values, so no imputation is needed. Outliers are kept because they may correspond to real situations, such as long-distance deliveries or heavy loads.
A great-circle distance is calculated from the coordinates to check the recorded distances, but it is not used as a model input. Departure hour is kept as an integer between 0 and 23. A cyclic encoding using sine and cosine was also tested but did not improve prediction accuracy.

To avoid data leakage, variables only available after a delivery, such as observed average speed, actual travel time and delay risk, are excluded from the models used for optimisation. These variables would not be available to a dispatcher when planning a delivery.
The 100,000 records are randomly split into training, validation and test sets using a fixed random seed of 42. The test set is kept separate during model development and is used for the final evaluation.

4 Machine Learning Models

Five regressors are trained on the same pre-dispatch feature set [3, 4]:

- Dummy regressor that always predicts the training-set mean (baseline);
- Ridge regression on the raw features;
- Ridge regression with engineered interaction terms;
- Random Forest;
- Histogram-based Gradient Boosting.

Categorical variables are one-hot encoded; continuous variables are standardised inside a scikit-learn pipeline.

Three standard metrics are reported [5]:

$$\mathrm{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|,$$
$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2},$$
$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

Table 1 summarises validation and test results.

Table 1: Model performance on validation and test sets (pre-dispatch features).

Model	val MAE	val RMSE	val R^2	test MAE	test RMSE	test R^2
Ridge + interactions	2.88×10^{-6}	4.21×10^{-6}	1.000	2.92×10^{-6}	4.26×10^{-6}	1.000
Hist. Gradient Boosting	2.40	3.20	0.9997	2.41	3.22	0.9997
Random Forest	4.07	5.85	0.9992	4.09	5.87	0.9991
Ridge (raw features)	32.7	44.2	0.952	32.5	43.9	0.952
Dummy (train mean)	167	201	≈ 0	167	201	≈ 0

Ridge regression with interaction terms achieves a test R^2 of 1.000 and an RMSE of 4.26×10^{-6}. The tree-based models also achieve high R^2 values, although their prediction errors remain larger. The near-perfect fit of the Ridge model may reflect simple mathematical relationships in the dataset and should not be interpreted as evidence of similar performance on real-world data. Histogram Gradient Boosting is retained as an independent audit model to assess the optimisation results.

5 Model Interpretation and Diagnostics

Permutation importance on the validation set ranks distance, cargo weight and the distance×cargo interaction far above every other variable (Figure 3). Vehicle type, road type and hour of day contribute essentially nothing once the continuous drivers are controlled for.

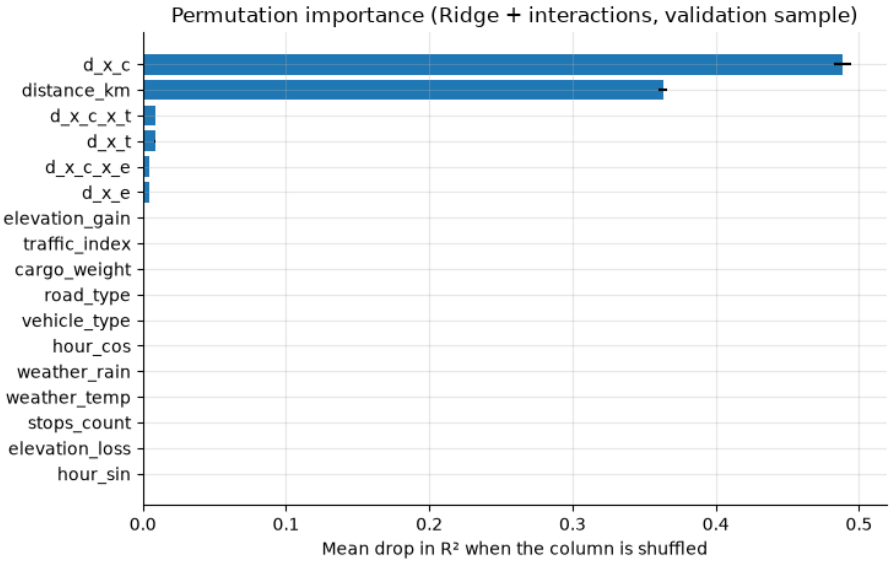


Figure 3: Permutation importance (drop in R^2) on the validation split for the selected model.

A comparison of the models provides further evidence that these variables add little predictive value. When vehicle type, road type and departure hour are added to the model, the validation RMSE increases slightly from 3.18 to 3.20. The R^2 obtained by using these three variables to predict the residuals of the model without them is -0.00033, indicating no improvement over a simple mean-based prediction.

On the test set, the interaction Ridge model has a mean residual of 1.6×10^{-8} and a standard deviation of 4.3×10^{-6}. An examination of the fitted coefficients suggests that CO_2 emissions follow an approximately multiplicative relationship of the form

$$CO_2 \approx \text{distance} \times f(\text{cargo}) \times g(\text{traffic}, \text{elevation}).$$

This relationship is consistent with the model's near-perfect predictions and helps explain why vehicle type, road type and departure hour contribute little additional predictive information in this dataset.

6 Emission Optimization (Problem A)

6.1 Decision space and constraints

For each of 400 randomly selected test deliveries the optimiser enumerates a grid of 1,728 candidate configurations (vehicle type $\times$ road type $\times$ hour $\times$ discrete speed levels). Travel time of each candidate is obtained from the exact identity discovered earlier. A candidate is feasible only if its travel time does not exceed $(1+s)$ times the recorded baseline time, with default slack $s = 0.10$.

Because the Ridge model is accurate to roughly 10^{-6}, a tolerance $\varepsilon = 10^{-6}$ is imposed: a candidate is preferred to the baseline only when its predicted emission is lower by at least ε. The value of ε is chosen on the validation scenarios and then frozen for the test evaluation.

6.2 Results

All 400 test scenarios have at least one feasible candidate. With $\varepsilon = 10^{-6}$, the optimiser changes the selected decision for none of the deliveries. Mean predicted CO_2 emissions remain at 296.31 before and after optimisation, corresponding to a mean reduction of 0%. The maximum reduction is also 0%. The audit model (Histogram Gradient Boosting) likewise reports no reduction.

When the tolerance is set to $\varepsilon = 0$, the optimiser changes the decisions for all deliveries, but the audit model records a mean reduction of -0.007%, indicating a very small increase in emissions. This suggests that removing the tolerance leads the optimiser to select alternatives based on negligible differences in predicted emissions, rather than meaningful improvements. This behaviour is consistent with the optimiser's curse [6].

These results are consistent with the earlier diagnostics. Within the tested candidate grid, the model finds no alternative that reduces predicted emissions by at least the chosen tolerance while satisfying the travel-time constraint. However, this does not establish that the original decisions are globally optimal or that vehicle and route choices have no effect on emissions in real-world delivery operations.

7 Traffic Scenarios and Sensitivity Analysis (Problem B)

A second, separate analysis asks a different question. Suppose a dispatcher is offered an alternative route that reduces the traffic index by a fixed amount $\delta \in \{0.10, 0.20, 0.30\}$ while possibly lengthening the distance by a relative factor x. How large may x become before the predicted emission of the alternative exceeds that of the original route?

These alternatives are purely hypothetical: they do not correspond to actual paths in a road network, elevation is held constant, and the traffic reduction is applied uniformly.

Empirically, the median break-even extra distance (at 10 % travel-time slack) is

Traffic reduction δ	Median break-even detour
0.10	1.5 %
0.20	3.0 %
0.30	5.0 %

Zero-detour predicted savings for the same three values of δ are 1.60 %, 3.01 % and 4.25 % respectively.

Varying the slack s over $\{-0.20, -0.10, 0, 0.10, 0.25, 0.50\}$ shows two clear patterns (Figure 4):

- In Problem A (vehicle/road/hour/speed) tightening or relaxing the time limit never changes the achievable emission level. There is no emissions–time trade-off visible in the data.
- In Problem B (traffic alternatives) the time constraint binds only when it is stricter than the recorded travel time. Once $s \geq 0$ the emission criterion alone determines the break-even detour.

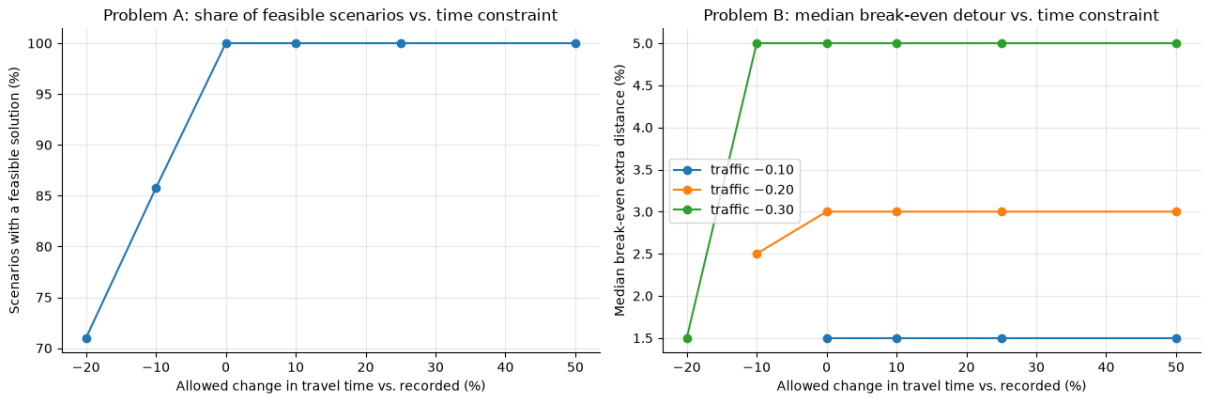


Figure 4: Sensitivity of Problem A feasibility (left) and Problem B median break-even detour (right) to the allowed change in travel time.

Stratification by cargo-weight and distance terciles yields essentially identical relative break-even percentages; only the absolute amount of CO_2 saved changes with the scale of the shipment.

8 Limitations and Discussion

Several limitations must be stated explicitly.

1. **Synthetic-data risk.** The near-deterministic emission formula, the impossible distance-versus-coordinate relationship, and the complete absence of signal in the decision variables strongly suggest that the file was generated by a mathematical model. Real fleet data exhibit measurement noise, driver behaviour, vehicle heterogeneity and network constraints that are missing here.

2. **Unknown semantics and units.** The meaning of the integer codes for vehicle and road type is undocumented. Units of CO_2, cargo weight and travel time are likewise unspecified.

3. **Missing operational constraints.** Vehicle capacity, driver hours-of-service, time windows, fleet availability, fuel cost and access restrictions are absent. The only constraint enforced is a relative travel-time slack chosen by the analyst.

4. **Hypothetical routes.** The lower-traffic alternatives examined in Problem B do not correspond to actual paths in a road network.

5. **Optimizer's curse.** When the tolerance ε is set to zero the search eagerly changes every decision, yet an independent model confirms that the changes are worthless. Any practical deployment must guard against exploiting residual model error.

The negative result on the controllable levers is therefore a statement about *this dataset*, not a claim that vehicle choice or departure timing never matter in the real world.

9 Conclusion and Future Work

9.1 What the experiments demonstrate

- A carefully engineered linear model can predict the CO_2 figures present in the file to machine precision from a handful of pre-dispatch features. The near-perfect accuracy is evidence of a simple underlying data-generating formula.
- The same model, when used inside a constrained combinatorial search over vehicle type, road type, departure hour and planned speed (Problem A), finds no emission-reducing configuration that respects a 10 % travel-time allowance.
- Emissions do respond to traffic exposure and distance (Problem B). The derived break-even detours supply a concrete, model-based rule that a dispatcher could apply *if* genuine lower-traffic route alternatives were available.

9.2 Future work

A natural next step is to replace the present file with real telematics data that include verified vehicle specifications, actual road-network paths, capacity and regulatory constraints, and multi-stop routes. Uncertainty quantification (quantile regression or conformal prediction intervals) would further protect against acting on differences smaller than the model's resolution. Any policy derived from historical data should be validated in a controlled pilot before large-scale deployment.

In summary, the study illustrates both the power and the limits of a pure predictive approach to emission optimisation: the model can be extraordinarily accurate, yet still yield no actionable savings when the controllable factors are independent of the target.

References

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